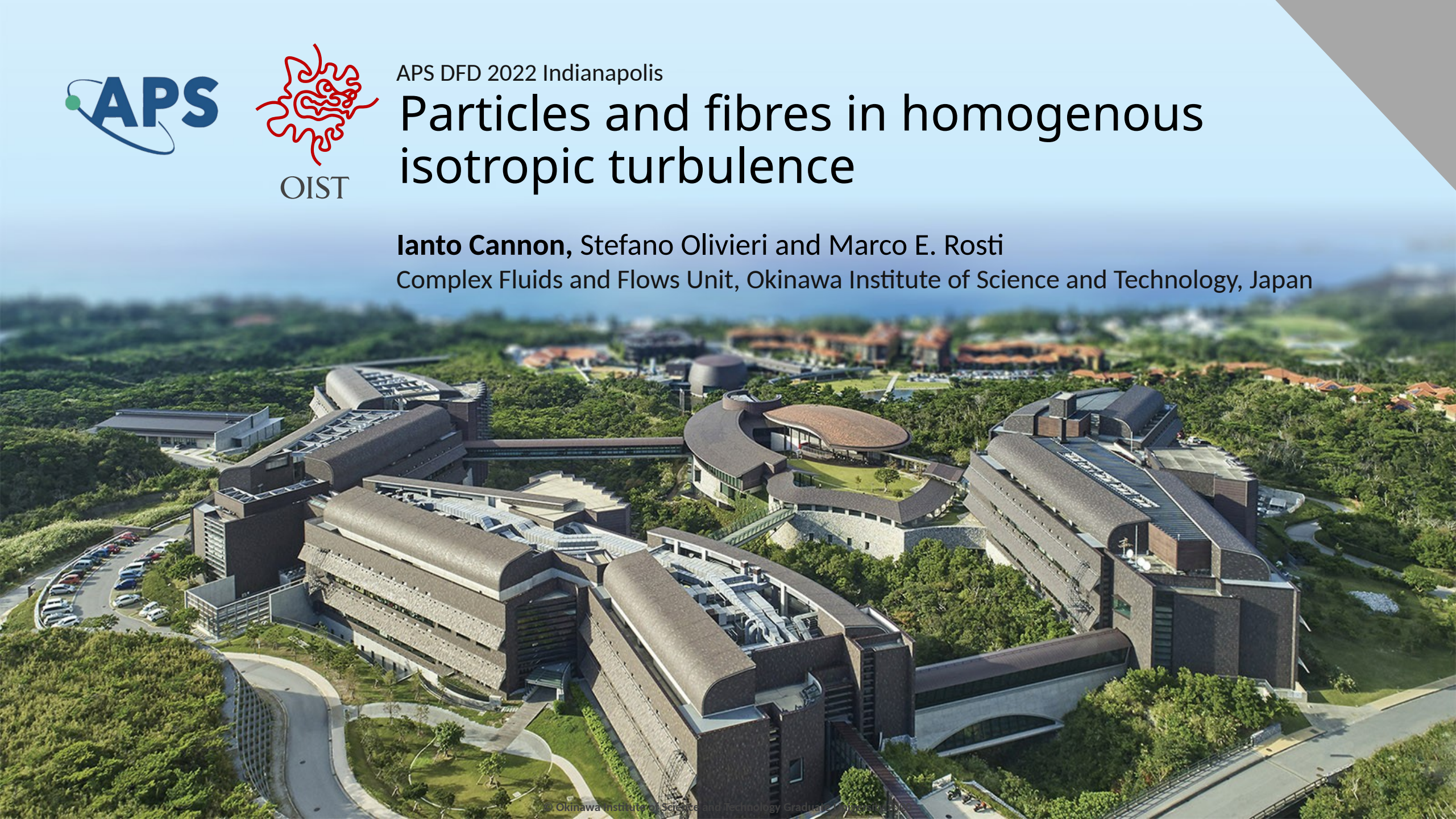
APS DFD 2022 Indianapolis

Particles and fibres in homogenous isotropic turbulence

Ianto Cannon, Stefano Olivieri and Marco E. Rosti
Complex Fluids and Flows Unit, Okinawa Institute of Science and Technology, Japan



Particles in **turbulent flows**

- Seen in many shapes and sizes in nature and industry

Microplastic dispersal in the ocean



Credit: Carmen Martinez Toron

Pollen dispersal in the atmosphere

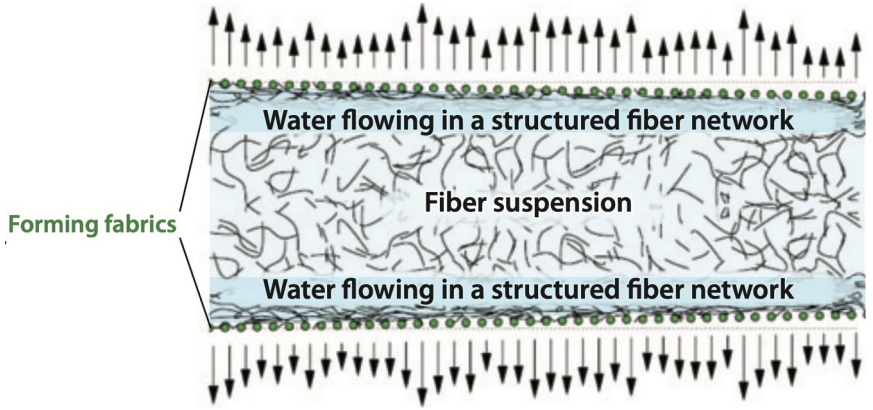


Credit: Robert J. Erwin

Pulp processing in papermaking



Credit: Toddmedia/Getty



Lundell *et al*, *Annu. Rev. Fluid Mech.* 2011

Background

- Previous studies (Voth & Soldati, 2017; Brandt & Coletti, *Annu. Rev. Fluid Mech.* 2021) often use one or more of the following assumptions:
  - small ($c \ll \eta$)
  - isotropic (spherical or point-like)
  - dilute (one-way coupling)

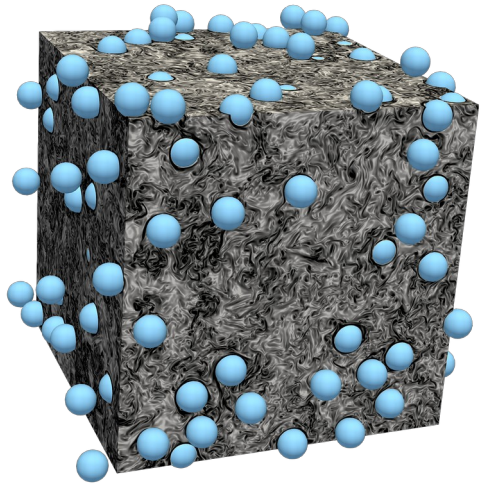


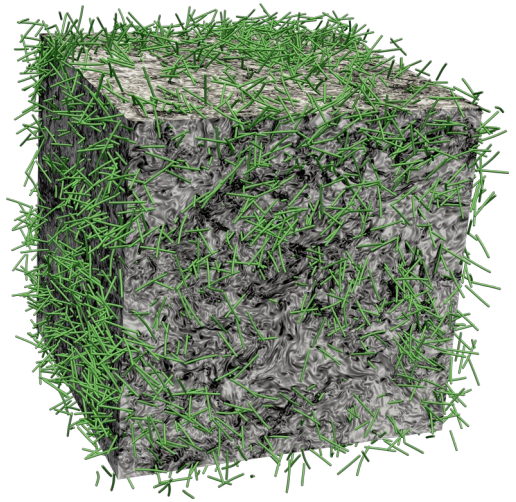


Goal of the work

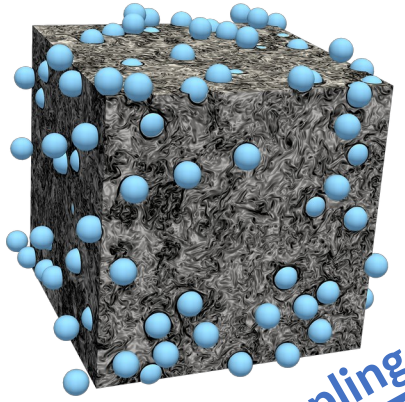
- We investigate particles which are:
  - finite size ($\eta < c < L$)
  - in non-dilute conditions (two-way coupling)
- We study the **turbulence modulation** by **isotropic** and **anisotropic** particles

How does particle **isotropy** affect the turbulent energy cascade?





Numerical method



- *N* **rigid spheres** with diameter c

$$m^p \frac{dU_i^c}{dt} = \oint_{\partial \mathcal{V}^p} \sigma_{ij} n_j dA + F_i^{\text{ext}},$$

$$\mathcal{I}^p \frac{d\Omega_i^c}{dt} = \oint_{\partial \mathcal{V}^p} \epsilon_{ijk} r_j \sigma_{kl} n_l dA + T_i^{\text{ext}}$$

Hori et al, Comput. & Fluids, 2021

Two-way coupling IBM

- Homogeneous isotropic turbulence (HIT) with $Re_\lambda \approx 435$

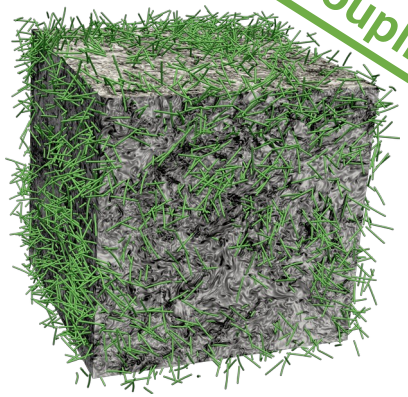
$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} = -\frac{1}{\rho_f} \nabla p + \nu \Delta \mathbf{u} + \mathbf{f}_{\text{FOR}} + \mathbf{f}_{\text{FS}}$$

$$\nabla \cdot \mathbf{u} = 0$$

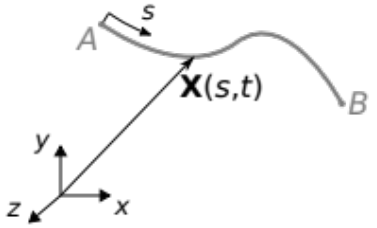
Two-way coupling IBM

- We vary the particle mass fraction

$$M \equiv \frac{m_p}{m_f + m_p}$$



- *N* **inextensible fibres** with length c

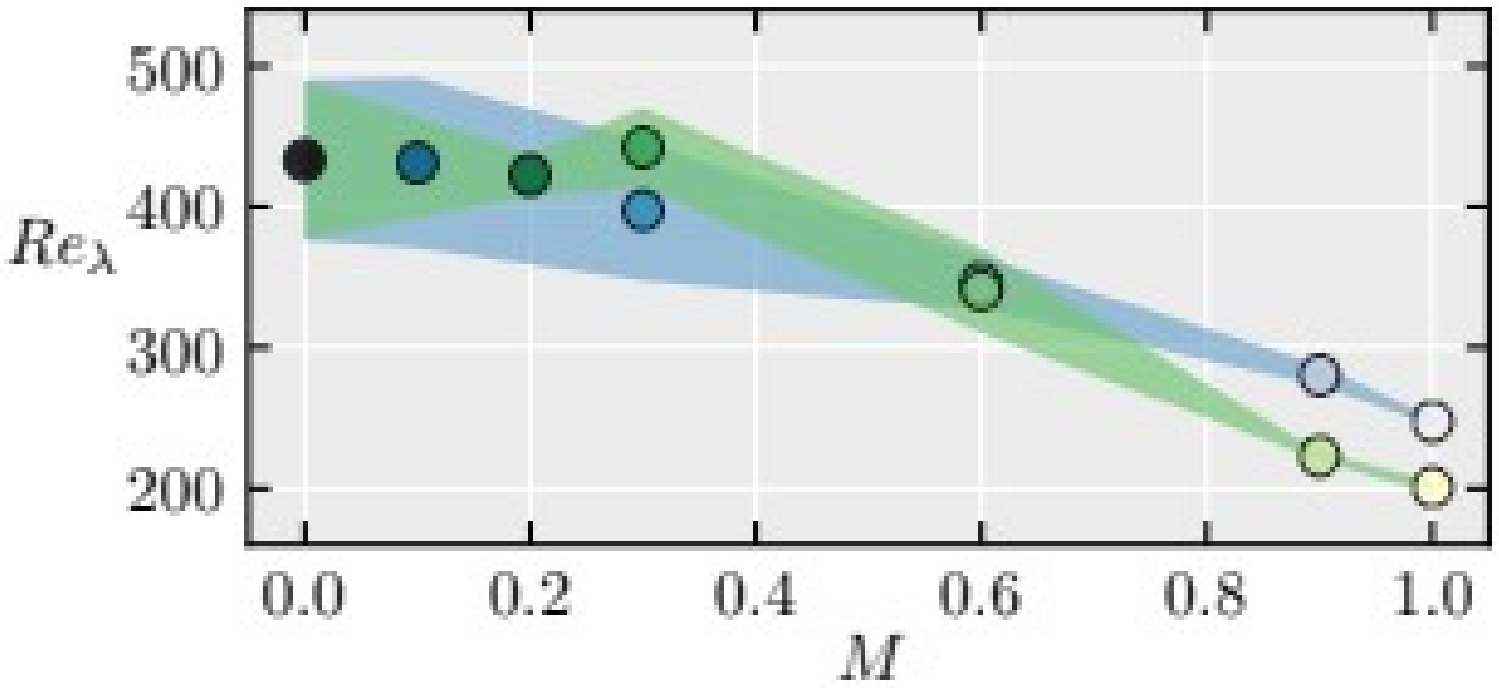


$$\frac{\partial \mathbf{X}}{\partial s} \cdot \frac{\partial \mathbf{X}}{\partial s} = 1$$

$$\Delta \tilde{\rho} \frac{\partial^2 \mathbf{X}}{\partial t^2} = \frac{\partial}{\partial s}\left(T \frac{\partial \mathbf{X}}{\partial s}\right) - \gamma \frac{\partial^4 \mathbf{X}}{\partial s^4} - \mathbf{F}$$

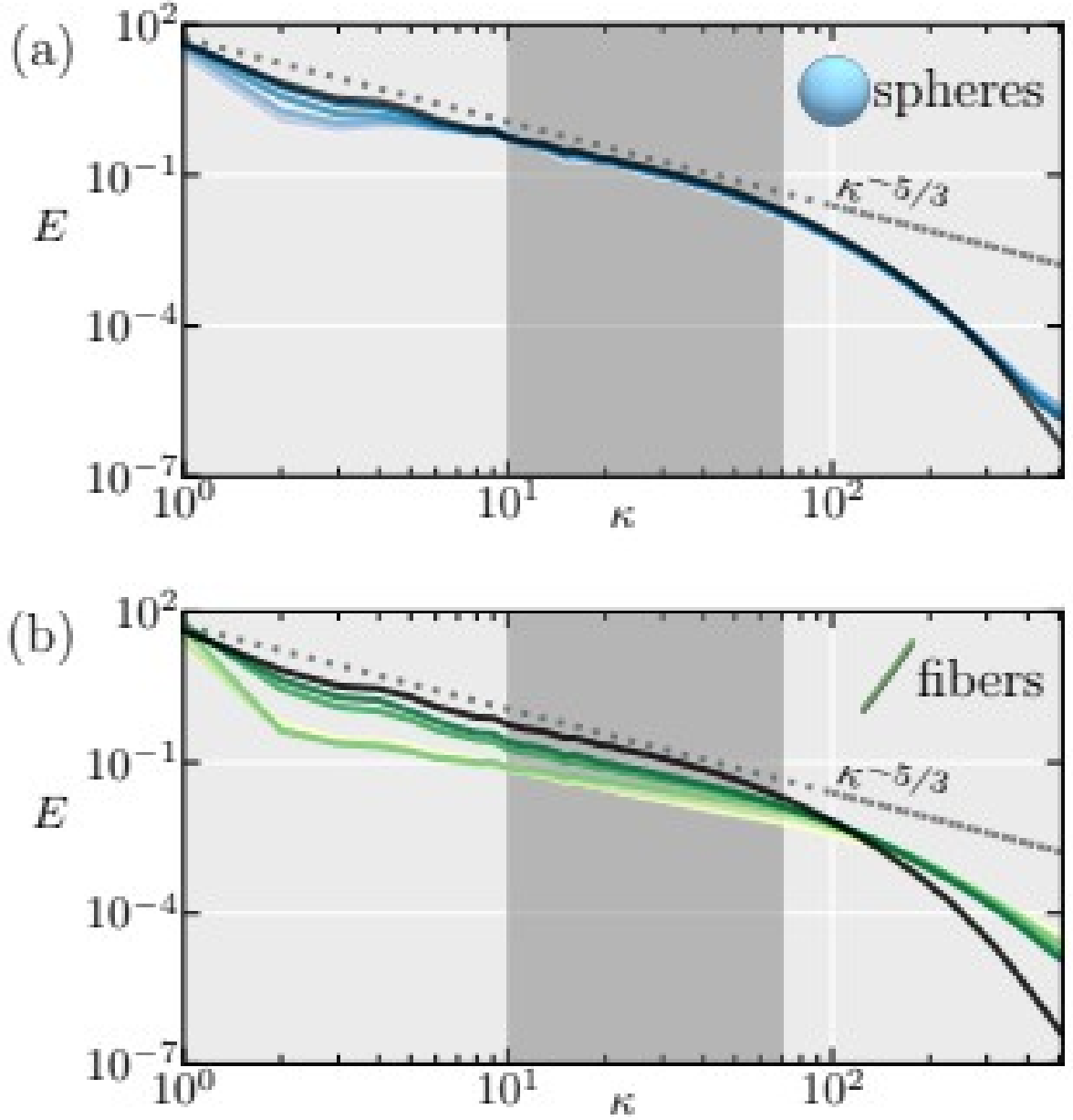
Olivieri et al, Phys. Rev. Lett., 2020
Huang et al, J. Comput. Phys, 2007

Effect of particles on bulk flow properties



Turbulence modulation

- Both spheres and fibres produce a similar reduction in the Taylor Reynolds number Re_λ.
- The flows with spheres show a reduction in the kinetic energy at the large scales.
- The fibres modulate the flow at all length scales



Kinetic energy scaling

- We fit power law functions $E = A\kappa^{-\beta}$ to the spectra

→ The flows with spheres retain the kolmogorov scaling

→ As the fibre mass fraction increases, the magnitude of the exponent β reduces

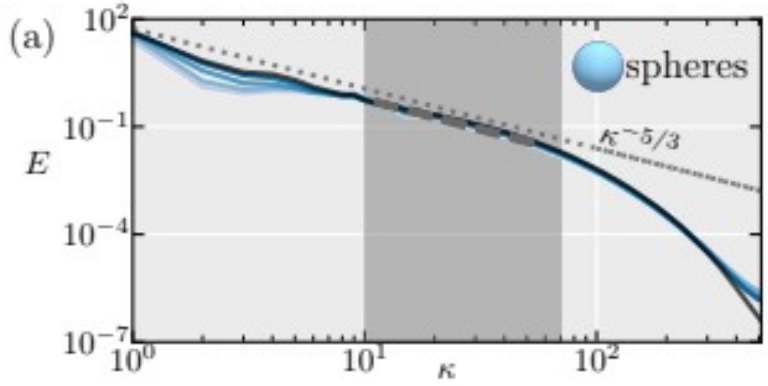
→This is because the fibres reduce the spatial correlation of the two point velocity difference $\delta u \equiv u' - u$.

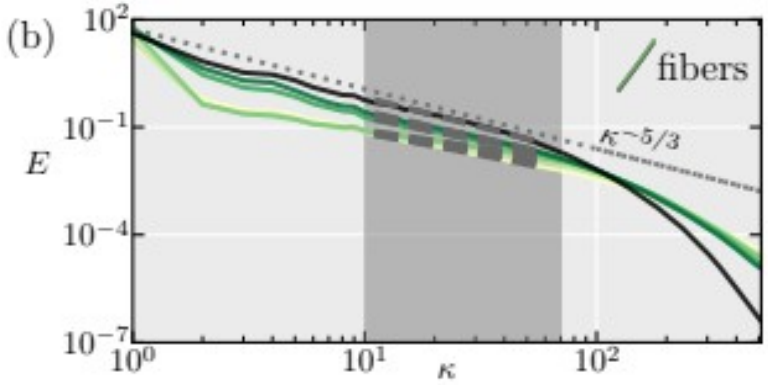


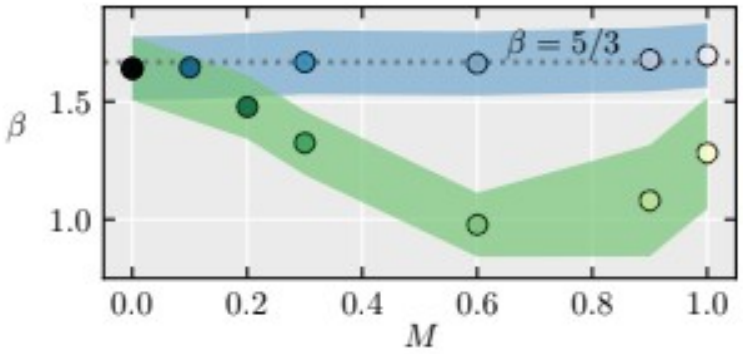
$$\langle (\delta u)^2 \rangle \sim r^{\gamma}$$

→Dimensional analysis gives us a relation between the energy spectrum and the velocity difference (Pope Cambridge University Press, 2000):

$$\gamma = \beta - 1$$





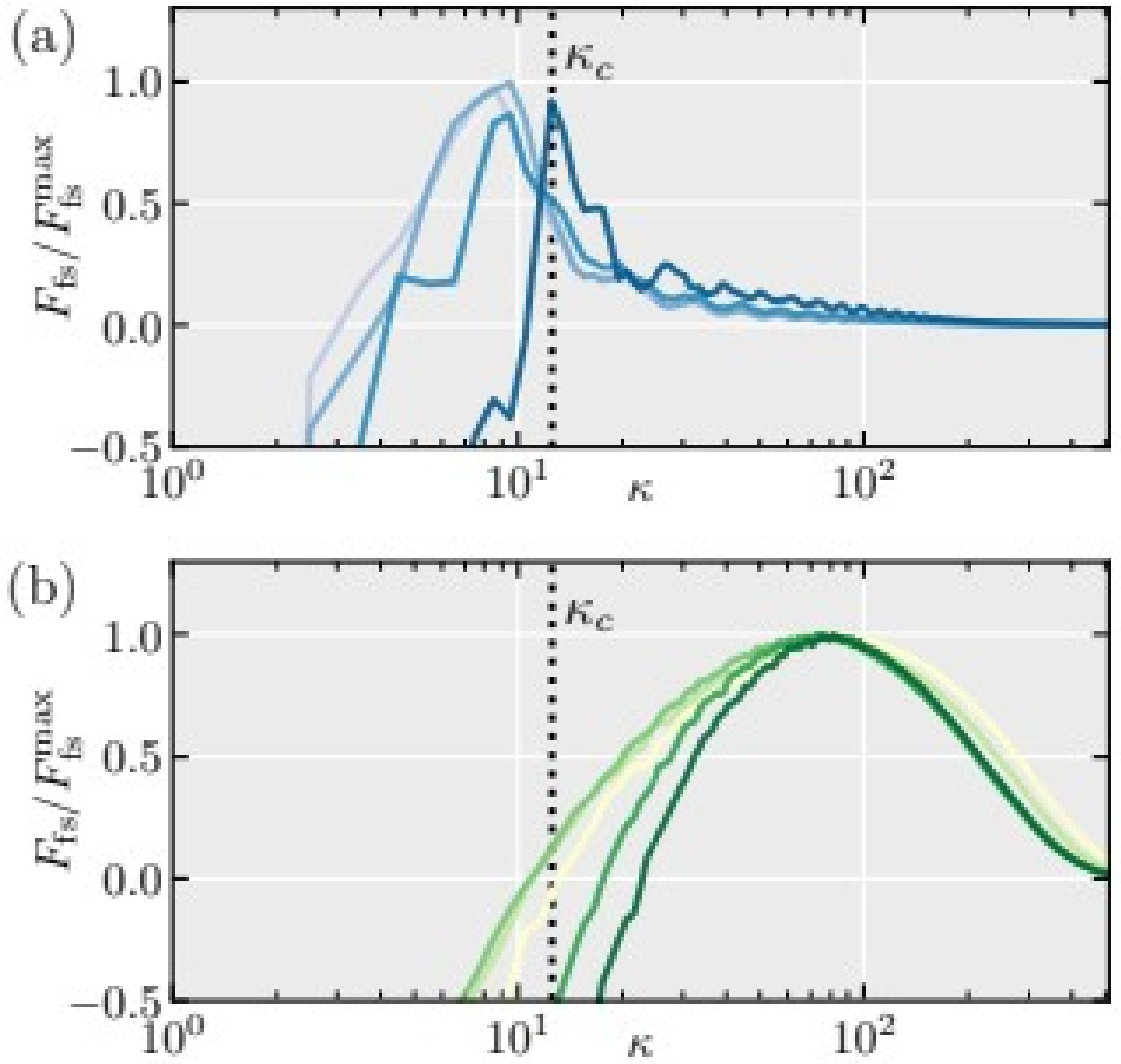


Fluid-solid coupling

- The spectrum of the fluid-solid coupling is

$$F_{\mathrm{fs}}(\boldsymbol{\kappa}) \equiv \frac{1}{2}(\hat{\boldsymbol{u}}^* \cdot \hat{\boldsymbol{f}}_{\mathbf{fs}} + \hat{\boldsymbol{u}} \cdot \hat{\boldsymbol{f}}^*_{\mathbf{fs}})$$

- We see the spheres act on the fluid at length scales comparable with their diameter c
- Fibres act on the fluid at a broad range of smaller scales



Conclusions

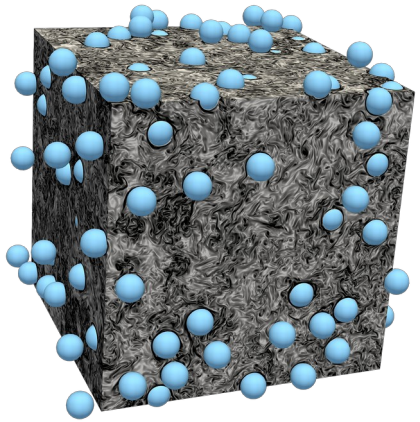
- We have focused on the modulation of turbulence due to *finite-size* particles, and, in particular, on the effect of their isotropy. We found:

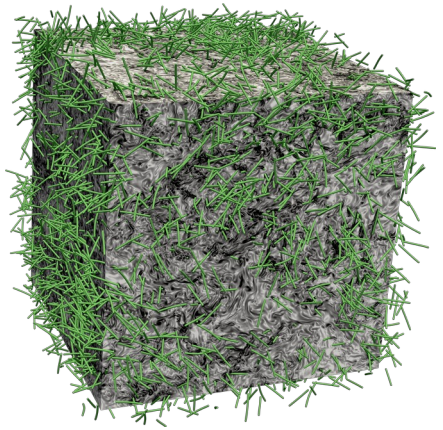
→The isotropic and anisotropic particles have a similar macroscopic effect on the flow (i.e., decrease of Re_λ)

→Isotropic particles act on the fluid at the length scale of their diameter. At smaller scales, the classical Kolmogorov cascade is unperturbed

→Anisotropic particles act on the fluid at a range of lengthscales, decorrelating the flow, and introducing a new scaling behaviour in the inertial range

Thank you!

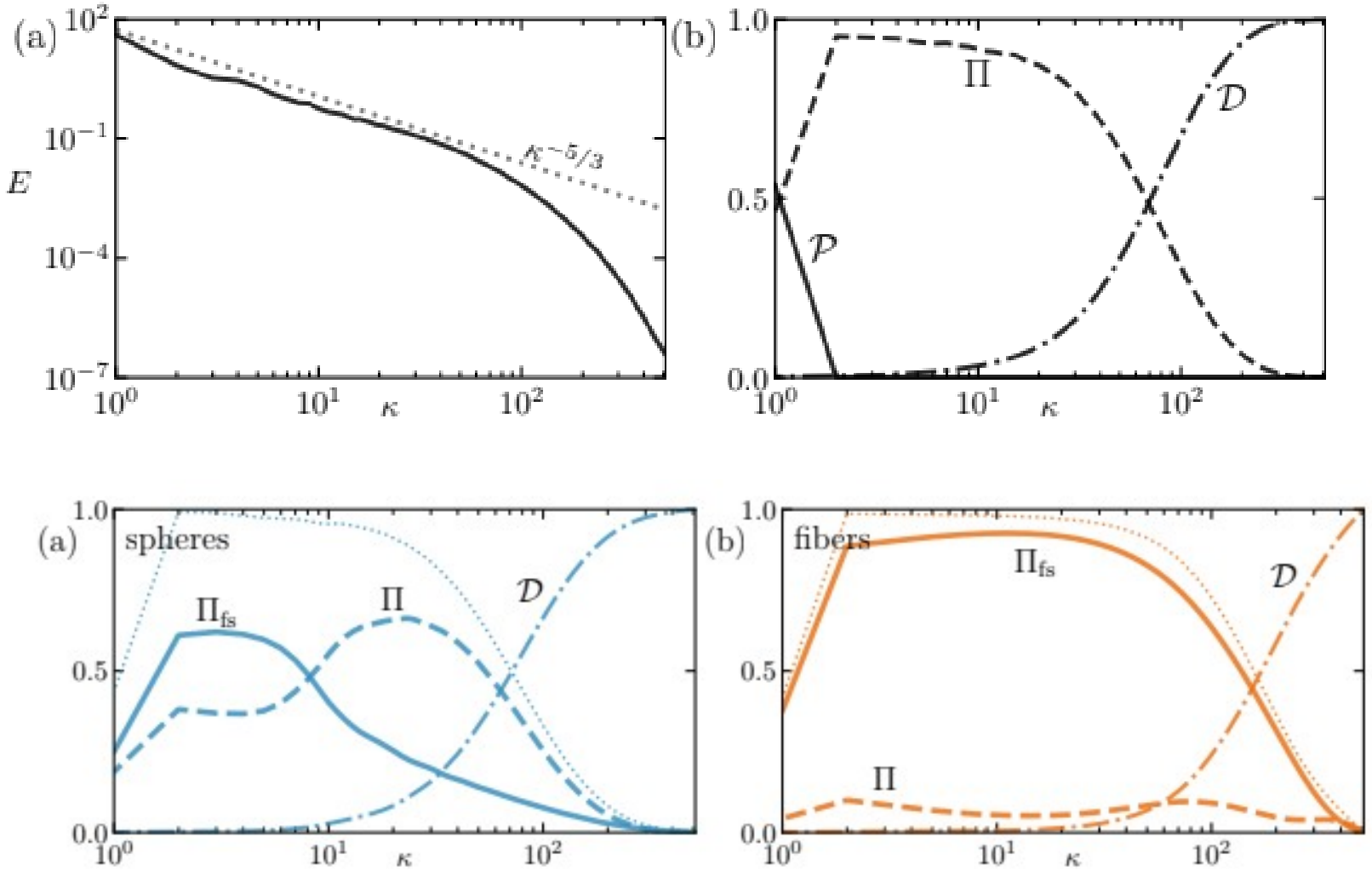




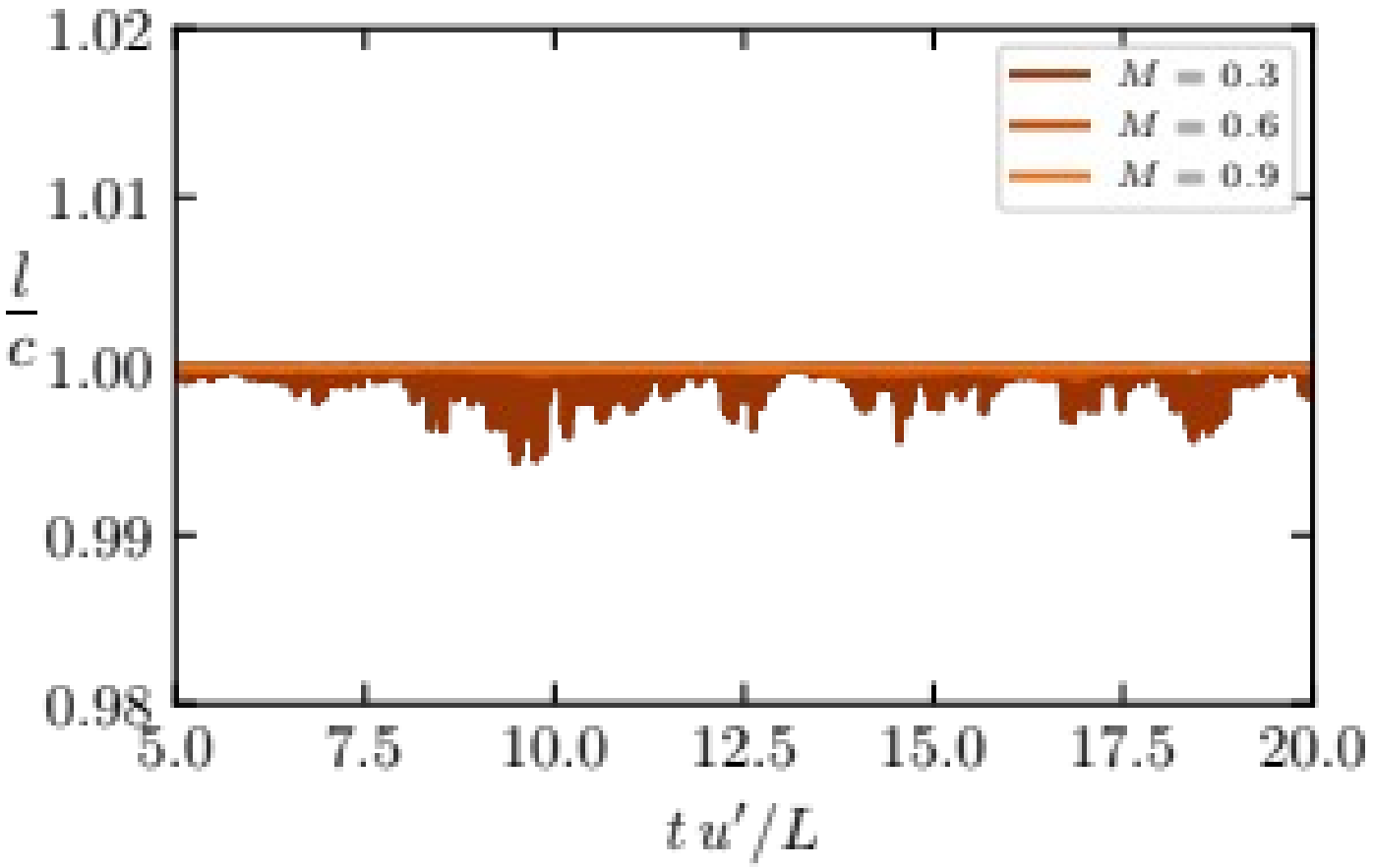
Olivieri, S., Cannon, I. & Rosti, M. E. (2022). The effect of particle anisotropy on the modulation of turbulent flows *(JFM Rapids)*

Energy fluxes

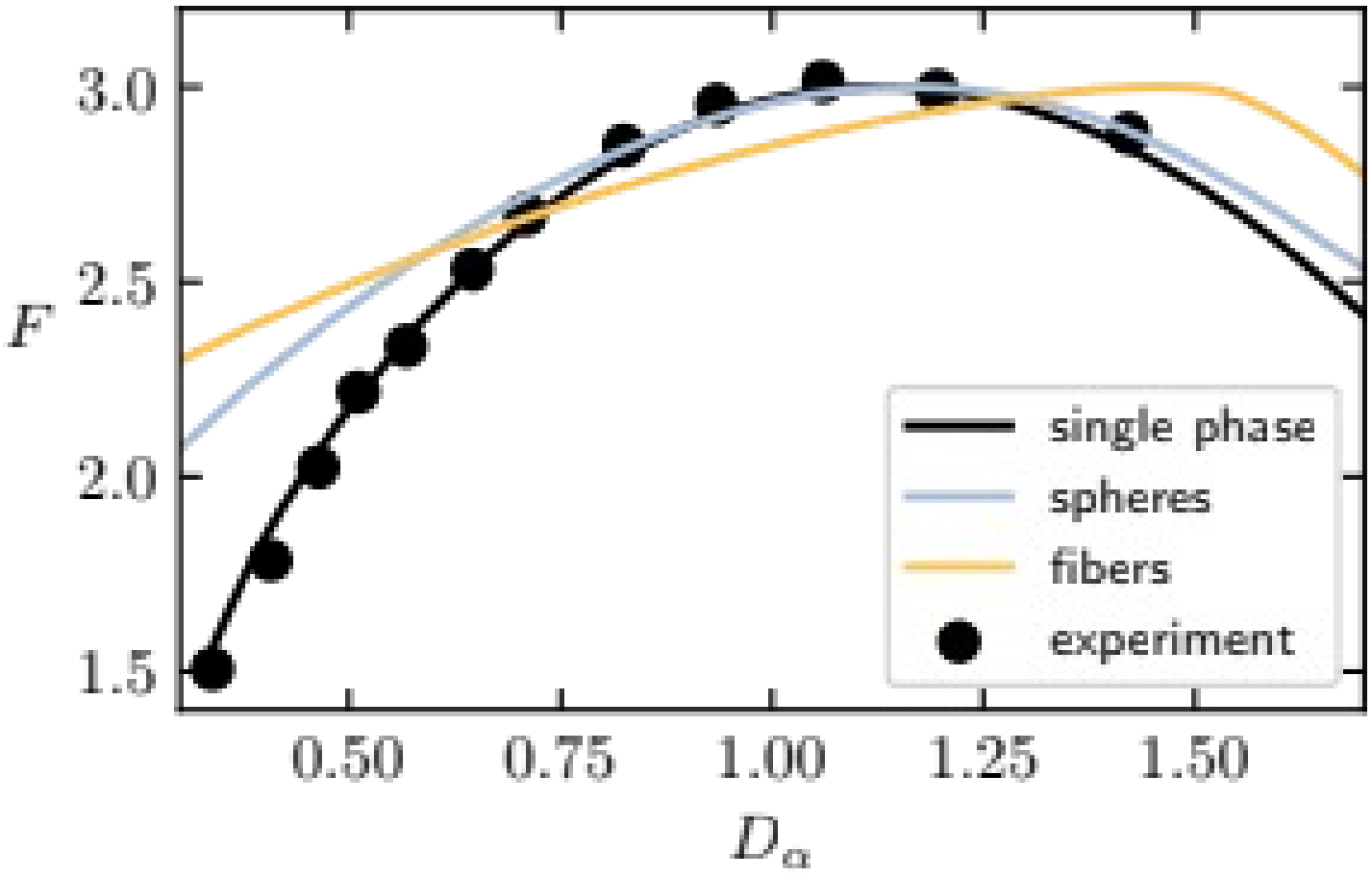
$$\mathcal{P}(\kappa) + \Pi(\kappa) + \Pi_{\mathrm{fs}}(\kappa) + \mathcal{D}(\kappa) = \epsilon,$$



Fibre deformation



Fractality of dissipation



I. Cannon: Particles and fibres in homogeneous isotropic turbulence